

# Sudharsun Lakshmi Narasimhan

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## Education

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| <b>Aalto University, Finland &amp; EURECOM, France</b><br><i>Erasmus Mundus Joint Master's Degree (Honors) in Security and Cloud Computing</i> | Aug 2023 – Sept 2025<br>CGPA: 4.74 /5 |
| <b>Amrita School of Engineering, India</b><br><i>B.Tech in Computer Science and Engineering</i>                                                | Jun 2016 – Jun 2020<br>CGPA: 8.68 /10 |

## Experience

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| <b>Security System Engineer</b><br><i>Ericsson HQ</i>                                                                                                                                                                                                                                                                                                                                                                                     | Kista, Sweden<br>Dec 2025 – Present         |
| <ul style="list-style-type: none"><li>Assessing risks and security posture of RAN compute and cloud infrastructure.</li></ul>                                                                                                                                                                                                                                                                                                             |                                             |
| <b>Master's Thesis Researcher</b><br><i>Ericsson Nomadic Lab</i> - Publication <a href="#">🔗</a>                                                                                                                                                                                                                                                                                                                                          | Kirkkonummi, Finland<br>Mar 2025 – Sep 2025 |
| <ul style="list-style-type: none"><li>Kernel-based Remote Attestation framework: eBPF system call tracing, TLS 1.3 packet analysis, Measured Boot, Trusted Platform Module.</li><li>Loadable Kernel Module (LKM): kfuncs, crypto APIs, character device; kernel-generated signing keys extended into TPM PCRs.</li><li>Temporal Convolutional Neural Network (TCN) for anomaly detection in system call sequences.</li></ul>              |                                             |
| <b>Security Research Trainee</b><br><i>Ericsson Nomadic Lab</i>                                                                                                                                                                                                                                                                                                                                                                           | Kirkkonummi, Finland<br>May 2024 – Aug 2024 |
| <ul style="list-style-type: none"><li>Elastic QUIC Reverse Proxy within Intel SGX enclave: EGo, Golang, Attested TLS, Azure Attestation.</li><li>Automated performance analysis: Python scripts to evaluate trade-offs between security and performance.</li></ul>                                                                                                                                                                        |                                             |
| <b>Teaching Assistant</b><br><i>Aalto University</i>                                                                                                                                                                                                                                                                                                                                                                                      | Espoo, Finland<br>Jan 2024 – Feb 2024       |
| <ul style="list-style-type: none"><li>Platform Security exercises: Evaluated and graded, providing constructive feedback to improve student comprehension.</li></ul>                                                                                                                                                                                                                                                                      |                                             |
| <b>Network Consulting Engineer</b><br><i>Cisco Systems</i>                                                                                                                                                                                                                                                                                                                                                                                | Bengaluru, India<br>Jan 2020 – Aug 2023     |
| <ul style="list-style-type: none"><li>5G Packet Core Automation: Developed a web application for deploying and managing service provider networks.</li><li>Kubernetes &amp; Containerized network functions: Hands-on experience with 5G core deployments.</li><li>Python scripting: Automated network configuration and monitoring tasks.</li><li>Network Automation Mentorship: Guided junior team members on best practices.</li></ul> |                                             |

## Technical Skills

**Areas of Interest:** Confidential Computing, Network Security, Software Development  
**Languages and Frameworks:** C, C++, Python, React, Django, SQL, MongoDB  
**Tools and Technologies:** Remote Attestation, eBPF, 5G Networks, Wireshark, Git, Containerization

## Academic Projects

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| <b>eBPF-Based Memory Dump Tool</b><br><i>Co-Developed with Andrea Oliveri</i> <a href="#">🔗</a> <a href="#">Github</a> <a href="#">🔗</a>                                                  | Oct 2024 – Feb 2025 |
| <ul style="list-style-type: none"><li>eBPF-based Memory Forensics tool: Captured and analyzed system RAM; gained expertise in kernel internals, uprobes, and memory management.</li></ul> |                     |

## Publications

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| <b>Fabric Defect Detection Using YOLOv2 and YOLO v3 Tiny</b>                                                                                                                                                                                                                                          | Nov 2020 |
| <ul style="list-style-type: none"><li>Gather and annotate datasets from the textile industry to develop a deep learning neural network model using the YOLO Algorithm for locating and classifying defects in fabric images. <a href="#">10.1007/978-3-030-63467-4_15</a> <a href="#">🔗</a></li></ul> |          |

## Awards

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| <ul style="list-style-type: none"><li>Ericsson Cooperation and Collaboration Award from Principal Researcher for teamwork and contributions</li></ul>              | June 2025     |
| <ul style="list-style-type: none"><li>Aalto School of Science, Dean's Incentive Scholarship</li></ul>                                                              | November 2024 |
| <ul style="list-style-type: none"><li>Erasmus Mundus Joint Master Degrees (EMJMD) Scholarship</li></ul>                                                            | August 2023   |
| <ul style="list-style-type: none"><li>Connected Recognition from Cisco's Vice President of Customer Experience for contributions to the Mobility project</li></ul> | March 2022    |